

MD. HASAN IMON

Junior AI/ML Engineer

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SUMMARY

AI/ML Engineer with specialization in Generative AI, NLP, and Machine Learning, with expertise in Deep Learning and developing production-grade multi-agent systems, RAG pipelines, and tool-augmented AI architectures. Skilled in fine-tuning LLMs and applying NLP techniques such as transformers and language modeling to build robust ML/DL solutions. Experienced in FastAPI, Docker, and CI/CD for designing and deploying scalable, end-to-end AI systems that enhance decision-making, drive automation, and deliver measurable business outcomes.

EXPERIENCE

Codixel

Junior Machine Learning Engineer

Jan 2026 – Present

Dhaka, Bangladesh

- Developed and fine-tuned Generative AI and machine learning models using transformer-based architectures, optimizing hyperparameters, evaluation metrics, and validation pipelines to improve model performance.
- Built AI-centric modular pipelines and LLM-powered applications including RAG and multi-agent workflows, collaborating with senior engineers to deploy scalable AI systems using FastAPI, Docker, and CI/CD pipelines.

Hi-Tech Parks

Machine Learning Engineer Intern

Oct 2025 – Dec 2025

Dhaka, Bangladesh

- Developed Retrieval-Augmented Generation (RAG) pipelines and multi-agent AI applications, collaborating with MLOps teams to deploy models using FastAPI, Docker, and CI/CD workflows.
- Performed data preprocessing, feature engineering, and model optimization to build reliable and scalable machine learning pipelines.

PROJECTS

MediGenius – Multi-Agent Medical AI Assistant

Github 🌐

- Built an AI-powered medical assistant using multi-agent orchestration to deliver context-aware healthcare consultations, achieving 82% clinical alignment, reducing misinformation by 60%, and improving query resolution rate by 75%.
- Architected a modular Fallback Agents with medical RAG over verified PDFs, SQLite with long-term memory, and a FastAPI backend with React JS frontend deployed via Dockerized CI/CD pipeline.

InformaTruth – AI Fake News Detection System

Github 🌐

- Built an explainable fake-news detection system using RoBERTa fine-tuning and multi-agent fact-verification pipelines. Implemented LangGraph orchestration with RAG retrieval, fallback agents, and FastAPI deployment pipeline with react.
- Boosted business impact and reliability by integrating memory, multi-agent tool routing, and fallback workflows. Achieved 70% accuracy and F1 score of 69% on the LIAR dataset with automated explanation generation.

- Developed a multi-agent document search system supporting PDFs, DOCX, TXT, and URL ingestion. Implemented vector-based semantic search with RAG architecture and dynamic tool routing.
- Stateful multi-agent workflow with dynamic tool routing, supporting ingestion, retrieval, LLM answering, and fallback agents. Improves query processing efficiency by 40% while handling 50+ simultaneous documents with Web apps.

TrueWealth AI – Multi-Agent Financial Advisor

- Built an AI financial advisor using agentic RAG pipelines with Query breakdown and multi-stage planning for real-time financial insights and decision support. deployed via Docker, FastAPI and React Web UI
- Achieving 99% query coverage with <3s P95 latency, reducing manual research effort for clients. Integrated financial news APIs and web search tools to provide reliable investment insights.

SKILLS

- **Programming & Data:** Python, Pandas, NumPy, Matplotlib, Seaborn, Data Analytics
- **MLOps & Backend:** FastAPI, Flask, REST APIs, Git, GitHub, Docker, OOP, Postman
- **Machine Learning & Deep Learning:** Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, PyTorch, TensorFlow, Keras, Scikit-learn
- **NLP & Large Language Models:** Transformers, Hugging Face, spaCy, NLTK, Text Classification, Embeddings, GPT, LLaMA, Mistral, Claude, PEFT, LoRA, QLoRA
- **AI Agents:** Multi-Agent Systems, LangChain, LangGraph, AgentOps, CrewAI, LlamaIndex, Agentic RAG, Tool-Augmented Agents, Memory-Augmented Agents, Prompt Engineering
- **Vector Databases:** FAISS, ChromaDB
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Cloud & Deployment:** Docker, CI/CD Pipelines, Render

EDUCATION

City University, Bangladesh

2022 – 2025

Bachelor of Computer Science and Engineering

CERTIFICATIONS

- **Machine Learning Specialization** Stanford University – (Coursera) 
- **Programming for Everybody** University of Michigan – (Coursera) 
- **Google UX Design Specialization** Google – (Coursera) 
- **Google IT Support Specialization** Google – (Coursera) 
- **IBM Cybersecurity Analyst Specialization** IBM – (Coursera) 